Functions and Function Notation

ANSWER KEY

Section 1: Evaluating Functions and Function Notation

1 Given $f(x) = 3x^2 - 4x + 1$, evaluate $f(-2)$.

a. 21 ✓

b. 5

c. -3

d. 17

$f(-2) = 3(-2)^2 - 4(-2) + 1 = 3(4) + 8 + 1 = 12 + 8 + 1 = 21.$

2 If $g(x) = \frac{2x+5}{x-1}$, find $g(3a + 1)$.

a. $\frac{6a+7}{3a}$ ✓

b. $\frac{6a+5}{3a+1}$

c. $\frac{6a+2}{3a}$

d. $\frac{2a+5}{a}$

$g(3a+1) = \frac{2(3a+1)+5}{(3a+1)-1} = \frac{6a+2+5}{3a} = \frac{6a+7}{3a}.$

3 Find all values of x for which $f(x) = x^2 - 6x$ equals -8.

a. $x = 2$ **and** $x = 4$ ✓

b. $x = -2$ and $x = -4$

c. $x = 1$ and $x = 8$

d. $x = 0$ and $x = 6$

$x^2 - 6x = -8 \Rightarrow x^2 - 6x + 8 = 0 \Rightarrow (x-2)(x-4) = 0 \Rightarrow x = 2$ or $x = 4.$

Section 2: The Vertical Line Test and Relations

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4 Which of the following equations defines y as a function of x?

a. $y = 2x^3 - 5$ ✓

b. $x^2 + y^2 = 16$

c. $x = y^2 + 1$

d. $|y| = 3x$

$y = 2x^3 - 5$ assigns exactly one unique output y for each input x. Equations with y^2 or $|y|$ can produce two outputs for a single x.

5 What does the Vertical Line Test determine about a graph?

a. **Whether the graph represents a function** ✓

b. Whether the function is one-to-one

c. The domain of the function

d. The slope of the curve

If every vertical line intersects a graph in at most one point, then the graph represents y as a function of x.

6 Which set of ordered pairs represents a function?

a. $\{(1, 2), (2, 3), (3, 4), (4, 4)\}$ ✓

b. $\{(1, 2), (1, 3), (2, 4)\}$

c. $\{(0, 5), (0, -5), (1, 2)\}$

d. $\{(3, 1), (3, 2), (3, 3)\}$

In $\{(1, 2), (2, 3), (3, 4), (4, 4)\}$, every first coordinate (x) is unique and paired with exactly one second coordinate (y).

Section 3: Operations and Difference Quotients

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7 Simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x)=4x-7$.

a. 4 ✓

b. $4h$

c. $4x+4h-7$

d. 0

$\frac{[4(x+h)-7]-[4x-7]}{h}=\frac{4x+4h-7-4x+7}{h}=\frac{4h}{h}=4.$

8 Simplify $\frac{f(x+h)-f(x)}{h}$ for $f(x)=2x^2+3$.

a. $4x+2h$ ✓

b. $4x$

c. $2x+2h$

d. $4x+h$

$\frac{2(x+h)^2+3-(2x^2+3)}{h}=\frac{2(x^2+2xh+h^2)-2x^2}{h}=\frac{4xh+2h^2}{h}=4x+2h.$

9 If $f(x)=x^2-1$ and $g(x)=2x+3$, find $(f\cdot g)(2)$.

a. 21 ✓

b. 15

c. 18

d. 28

$f(2)=2^2-1=3.\ g(2)=2(2)+3=7.\ (f\cdot g)(2)=f(2)\cdot g(2)=3\cdot 7=21.$

Section 4: Function Composition

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10 Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(x)$.

a. $2x^2 - 5$ ✓

b. $4x^2 + 4x - 2$

c. $2x^2 - 3$

d. $2x^2 + 1$

$(f \circ g)(x) = f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.

11 For the same functions $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, evaluate $(g \circ f)(2)$.

a. 22 ✓

b. 25

c. 17

d. 3

$f(2) = 2(2) + 1 = 5$. Then $g(f(2)) = g(5) = 5^2 - 3 = 25 - 3 = 22$.

12 If $h(x) = \sqrt{2x^3 - 5}$, find functions $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$.

a. $f(x) = \sqrt{x}, g(x) = 2x^3 - 5$ ✓

b. $f(x) = 2x^3 - 5, g(x) = \sqrt{x}$

c. $f(x) = x^3, g(x) = \sqrt{2x - 5}$

d. $f(x) = 2\sqrt{x}, g(x) = x^3 - 5$

Outer function is the square root: $f(u) = \sqrt{u}$. Inner function is the cubic polynomial: $g(x) = 2x^3 - 5$.

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